AUTO SPECIAL

COLUMN



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NEWEST VISION
BMW goes retro with its Neue Klasse range of EVs; below, Nissan's solid-state battery (left) is nearly half the size and weight of its current battery (right)

THE EVs OF THE FUTURE

THE DESIGN AND TECHNOLOGICAL WONDERS THAT AWAIT US IN THE FUTURISTIC WORLD OF ELECTRIC CARS

While it has taken Internal Combustion Engine (ICE) cars all of about 150 years to get to the current stage of development, electric vehicle (EV) development, by comparison, is still in its infancy. Even though the early EV made its appearance before the ICE vehicle, the rapid development and practicality of the ICE vehicle meant that EVs faded into near-oblivion. The real EV revolution started just over a decade back, as vehicles like the Nissan Leaf followed by a slew of EVs from the likes of BMW, Renault, Tesla and others presented a viable alternative.

Now, after a decade and a half of rapid development on EVs and with all global firms devoting most of their R&D funds to them, the electric vehicle space is poised for a quantum leap in all aspects, whether the shape and form, or electric motors or the digital experience or the storage of electricity.

First and foremost, all companies are deciding what their vehicles will look like. While some were pursuing design philosophies similar to their ICE vehicles, a radical shift seems to be en route. Some are going boxy and practical, others are going space-age or have video game looks and a few seem to be straight out of the pages of a manga. Yet others are going in for classic designs. A few are developing vehicles that will have a powertrain module and a passenger module and both can be changed by the owner in minutes, depending on the needs from the car. The vehicle can be changed from a van seating eight and their luggage to a lounge for two just by changing the passenger module.



The hardware that is going to run these cars is getting more powerful with many companies planning to equip each car with computing powers. As these cars develop systems that will keep them out of harm's way, the driving experience is going to transform and the driver will have the option to create her own reality inside the cockpit of the car and infotainment is going to be all-encompassing.

Motors are getting lighter and smaller, so the already non-obtrusive engine of the EV can be hidden away while at the same time battery packs will also be lighter, smaller and fast-charging.

And for those who want to experience the old-fashioned joy of driving, an ICE vehicle can be made to sound, feel and drive like an ICE sportscar! This is already possible in an electric vehicle like the Hyundai Ioniq 5N.

Much of this may sound utopian, but we will see all this in the next decades. Given the focus of the automotive industry on EVs and the entry of many new players, the entry barrier of 150 years of development is gone. Many new players are from different fields—like Sony, Apple, Google or Amazon. It will be interesting to see what they come up with in response to what the traditional players are planning! ■